

Insurance Premium Prediction Project

Day 2: Data Preprocessing & Pipeline Architecture - 2026-07-29

1. Executive Summary

Following Day 1's exploratory data analysis, Day 2 focuses on establishing a robust, leak-free preprocessing pipeline. The goal is to properly handle missing values, handle categorical features, and prepare the dataset for model training while rigorously avoiding data leakage.

2. Backup and Duplicate Checks

We began by creating a full backup of the raw dataframe. Then, we verified the uniqueness of the Property_ID and checked for duplicate rows. Maintaining a clean foundation ensures that our model will not be biased by over-represented, repeated observations.

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Account of नाम खाता		Debit नाम		Credit जमा		Dr. or Cr. लेनी देनी	Balance शेष	
Date तिथि	PARTICULARS विवरण	C.B. Folio शेकड पृष्ठ	₹	P.	₹	P.	₹	P.
	Nepacut 40	222			1212		10/25	
	Breazon 10	176			2 strip		7/25	
	Gilny - m 25	95			1 strip		8/25	
	1000 2000	450			1000		10/25	
							16/8/25	
13/09/25	E 1000	419			1 trial		9/25	
03/10/25	2 vaccine to	712			5 st		10/25	
	Veride 100	155			10 + 2 st		11/25	
	Promixol - 1	274			2 strips		12/25	
	Grabapin NT	500			1 strip		05/28 Brokage	
	Panthum DSR	177			1 strip		11/25	
	Orat 40 Am	354					10/25	
	Esofag - 40	174			1 strip		12/25	
	Kefarist - Plus	138			1 st		2/26. Brokage	
	Lamizal - 50	200			6 st		11/25	
	Bonmugita - P	211			2 st		12/25	
	Eglyduo - 5/100	176			1 st		12/25	
	Clabg - mt 10	110			3 st		12/25	
	Am 1000	249			3 st		11/25	
	Lipicure ts	419			2 st		11/25	
	Prazomil - 100	115			6 st		11/25	
	Skabazon	235			1 st		11/25	
							18/10/25	
13/12/26	Serolix - 25	50			10 st		1/26	
	Best - kala - 50	64			5 st		3/26	
	Serolix 100	130			11 st		1/26	
	CB - D3 Team	320			4 st		2/26	
	1000 2000	302			6 st		3/26	
	Grinkolam VP	295			10 st		3/26	
	Storay - 500	790			6 st		2/26	

Choice

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3. Analyzing Missing Values

We generated a missing value summary table. The output revealed that categorical columns like `Security_Level` and `Roof_Type`, alongside numerical columns like `Previous_Claim_Amount_INR` and `Credit_Score`, exhibit missing values.

LEDGER खाता

Account of नाम खाता 254

Date तिथि	PARTICULARS विवरण	C.B. Folio रोकड़ पृष्ठ	Debit नाम		Credit जमा		Dr. or Cr. लेनी देनी	Balance शेष
			₹	P.	₹	P.		
	Rafanex Pluxy	184		5st			1/26	
	Zonyl-0.5	110		7st			2/26	
	Dofac 30	190		2st			3/26	
	Casdec am 5	356		3st			4/26	
	Gabapin NT 300	308		1 strip			02/27 breakage	
	CycloReg	53		5 strips			04/26	
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4. Train-Test Split & Preventing Data Leakage

A critical step before any imputation or encoding is isolating a test set. By splitting the data into 80% training and 20% testing sets using an 80/20 split, we ensure that the test set remains unseen. Imputing missing values using the entire dataset's median would introduce 'data leakage', where information from the test set inadvertently influences the training phase.

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Account of नाम खाता		LEDGER खाता		385	
Date तिथि	PARTICULARS विवरण	C.B. Folio रोकड़ पृष्ठ	Debit नाम	Credit जमा	Dr. or Cr. लेनो देनो
			₹ P.	₹ P.	Balance जेब
	Fluxset 40	62			
	Fluxset 20	45			
	Fluxset 60	85			
	Sartel - Am 80	213			
	Silotime - m 50	550			
	Dapanel - m 5/1000	109			
	Targon - m	288			
	Melomerve Plus	168			
	Zupion - DP	120			
	Olomat 40	322			
	Simarest - AP	90			
	Pantium - DSR	172			
			10st		6/26
			10st		6/26
			9st		6/26
			1st		4/26
			5st		6/26
			2stnd		5/26
			1st		6/26
			3st		6/26
			11stnd		6/26
			1st		5/26
			3st		6/26
			1st		5/26
					4/26
2/5/26	Remiva. m-forte	404			
			2st		4/26

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5. Feature Separation

To properly construct a preprocessing pipeline, we divided the variables into numerical and categorical feature sets. Identifying feature types enables tailored transformations for each group.

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6. Scikit-Learn Preprocessing Pipeline

A professional machine learning pipeline was built using scikit-learn's Pipeline and ColumnTransformer. For numerical features, a SimpleImputer was configured with the 'median' strategy. For categorical features, a Pipeline was created chaining a SimpleImputer ('most_frequent' strategy) and a OneHotEncoder. This centralized preprocessor guarantees consistency and prevents data leakage across future deployments.

10/7/24	Amoxicillin 400	196	1st	10/24
	Clonidine 2.5	145	1st	10/24
	Singulair 400	391	1st	10/24
	C No.	000011	19/10/24	5/9/24
16/6/24	Netuz	196	3 stb	10/24
	Pregabalin 75/30	261	3 stb	10/24
	Gabapentin ME	307	1 st	10/24
	Azithromycin 40	155	5 stb	12/24
	Duvant 60	256	2 stb	12/24
	Atzil-M 10			12/24
	Parnon 13 SR	60	5 stb	11/24
	Cloze mt 10	98	1 st	11/24
	Cardace 10	407	1 st	4/26 - Benz
	Neelidiz	114	1 st	12/24
				16/11/24

Choice